

Project Initialization and Planning Phase

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PROJECT TITLE: DIABETIC RETINOPATHY CLASSIFICATION WEB APPLICATION

MAXIMUM MARKS: 3 MARKS

1. Project Objective

The objective of this project is to develop a web-based application that classifies Diabetic Retinopathy (DR) from retinal images using Deep Learning techniques.

The system aims to:

- Detect DR at early stages
 - Classify severity levels
 - Assist doctors in faster diagnosis
 - Reduce manual screening effort
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2. Scope of the Project

The scope includes:

- Image upload through web interface
 - Deep learning model prediction
 - Classification into DR stages:
 - No DR
 - Mild
 - Moderate
 - Severe
 - Proliferative DR
 - Display prediction results to users
 - Store prediction results in database
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3. Technology Stack

- **Frontend:** HTML, CSS
- **Backend:** Flask (Python)
- **Model:** CNN using TensorFlow/Keras
- **Database:** SQLite

- **Deployment Environment:** Localhost

4. Project Timeline

Phase	Task	Duration
Phase 1	Problem Definition & Research	1 Week
Phase 2	Data Collection & Preprocessing	2 Weeks
Phase 3	Model Development & Training	2 Weeks
Phase 4	Web App Development	1 Week
Phase 5	Testing & Optimization	1 Week
Phase 6	Documentation & Final Submission	1 Week

5. Expected Outcome

- A functional DR classification web application
- Accurate prediction using trained CNN model
- User-friendly interface
- Efficient screening support tool